

Phantom Brand Persistence

AI Presence Index v0.7 — Bed Bath & Beyond, designed for test

4 May 2026 · Pablo Ulpiano Gonzalez Castro · Third System

Independent measurement for the AI mediation layer.

AI knows the brand is closed. It surfaces the brand anyway.

In v0.6 we measured a single defunct brand — Mint, the personal finance application Intuit shut down in March 2024 — appearing in 44% of AI responses about live products 25 months later. We called it phantom brand presence. Phase 2 was designed to test whether the result was a property of AI mediation or an artefact of one category.

Third System measured Bed Bath & Beyond, a household-goods retailer that filed for bankruptcy in April 2023, liquidated its physical stores, and was acquired by Overstock — which rebranded itself to Beyond, Inc. and relaunched bedbathandbeyond.com as an online-only entity. We measured BBB across six pre-registered prompts, six frontier AI models from four labs, eight runs each — 288 successful measurements committed to a pre-registration document locked before any data was collected.

The phantom replicates. Bed Bath & Beyond surfaced in 38.2% of all AI responses about household goods retailers — comparable to Mint's 44% in v0.6 personal finance, in a different category, with a more rigorous methodology, against eight pre-registered hypotheses including a structural comparator (Pier 1, which collapsed earlier and shows zero phantom rate) the v0.6 measurement could not pull.

But the phantom does not replicate as we expected. In only 1.7% of measurements does AI present BBB as fully live with no caveat. In 88% of BBB-containing responses, AI discloses the closure, the rebrand, or the online-only status. The AI knows what happened. It surfaces the brand anyway, with disclosure.

This is a structurally different finding from the lag interpretation we offered for Mint. The phantom is not in AI's knowledge of the entity. The phantom is in AI's recommendation-set composition. Brand-mention pathways become reinforcement-encoded in AI mediation independent of entity status. The brand's accumulated editorial weight in the training corpus reserves a recommendation slot that the AI then fills with disclaimer, rather than vacating.

Five findings hold across the measurement. **First**, the phantom replicates: BBB surfaces in 38.2% of all AI responses, with a structural comparator (Pier 1, smaller pre-collapse footprint, longer time-since-collapse) at exactly 0.0% across every model and prompt.

Second, the phantom is a recommendation slot, not a knowledge gap. In 88% of BBB mentions, AI demonstrates knowledge that the entity has changed. The naive-phantom rate — AI presenting BBB as fully live with no caveat — is 1.7% of all measurements. The brand still surfaces in 38.2% of responses. That gap is where the structural finding lives.

Third, newer models phantom-mention better, not less. Within Anthropic, Opus 4.7 surfaces BBB more often than Sonnet 4.6 (56.2% vs 41.7%) but a higher fraction of Opus mentions are aware-mode. Within OpenAI, gpt-5.4-mini surfaces BBB more often than gpt-5.5 (50.0% vs 14.6%) but with a far higher aware-mode rate. The recommendation slot persists; the disclosure improves.

Fourth, different prompts activate different temporal frames. FUNCTIONAL prompts produce BBB-as-current-online-retailer (70.8% Presence, 60.4% caveated). IDENTITY prompts produce BBB-as-cultural-memory (81.2% Presence, 43.8% aware-mode). DISCOVERY prompts produce zero BBB mentions. The same brand occupies different temporal slots depending on the cognitive frame the prompt activates.

Fifth, the rebrand has not propagated. Of 110 BBB mentions, 0% reference Beyond, Inc. as a corporate parent in isolation. 75.5% reference only the legacy 'Bed Bath & Beyond' string. The new corporate identity captures only the disclosure-language slot; the legacy brand continues to drive AI visibility long after the corporate identity has changed.

Marketers operating in disrupted categories should expect their brand's AI-mediated visibility to outlast their entity's structural integrity, and consumers in those categories will encounter the brand-with-caveat, not the brand-as-warning. AI brand visibility is not a knowledge freshness problem. It is a recommendation-set composition problem, and retraining alone may not resolve it.

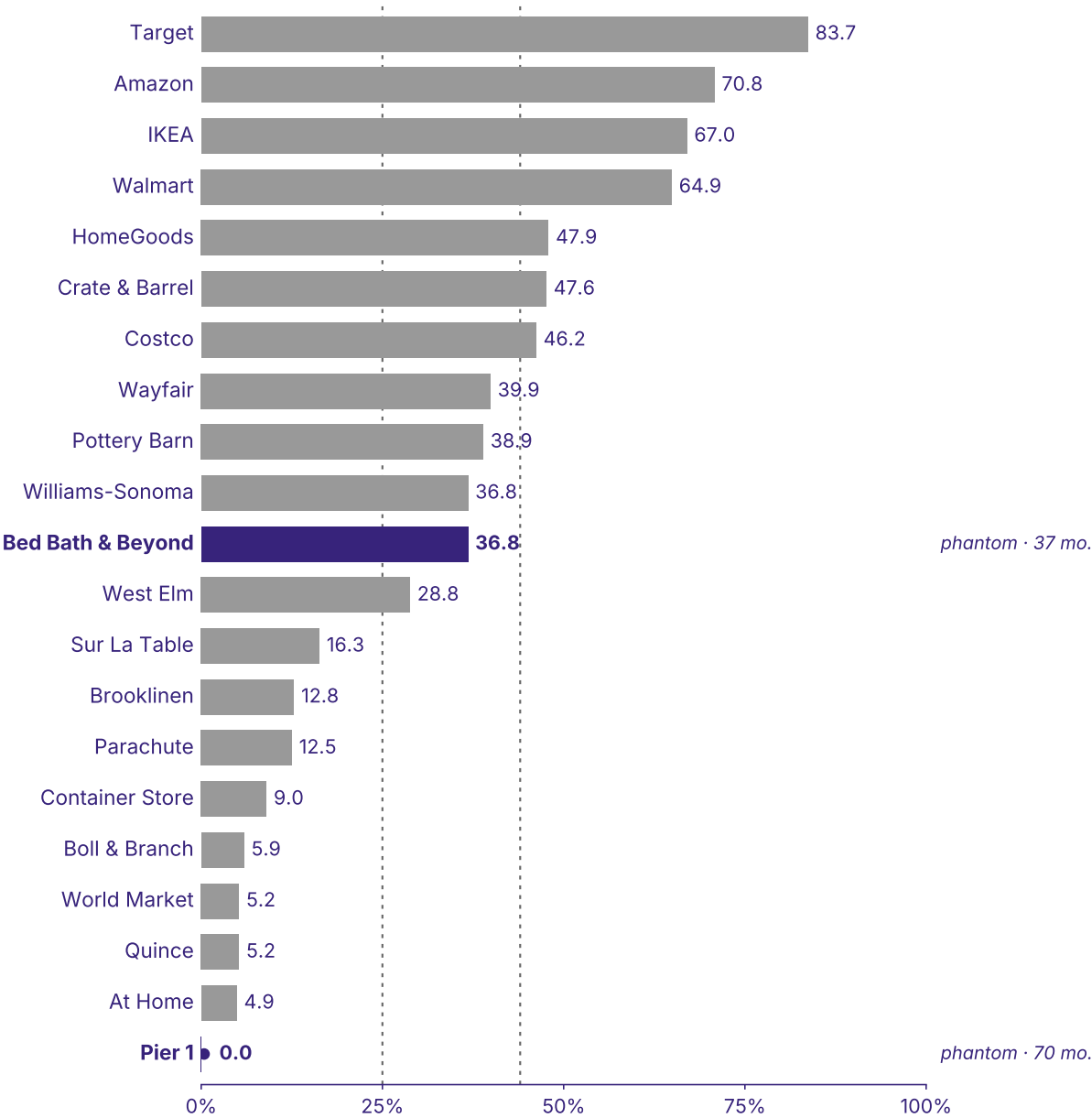
Household goods leaderboard

Twenty-one retailers ordered by AI Presence. Bed Bath & Beyond and Pier 1 are highlighted as phantom test subject and structural comparator — the foundation for the five findings that follow.

AI Presence — Household Goods Retail

21 brands ordered by Presence. Bed Bath & Beyond and Pier 1 highlighted as phantom and comparator.

H1 threshold Mint v0.6 (44%)



Source: Third System AI Presence Index v0.7 · n=288 measurements (6 prompts × 6 models × 8 runs) · 4 May 2026

Figure A · Twenty-one household-goods retailers by AI Presence. BBB at #11 (36.8%) sits between Williams-Sonoma and West Elm; Pier 1 at #21 (0.0%) appears in zero of 288 measurements. The phantom signal is structurally specific.

What we measured

Phase 2 follows the AIAS Presence Measurement Protocol v1.0, locked before the v0.6 cross-category findings were published and used unchanged for this measurement. The category was selected explicitly as a designed-for-test target for Pattern 6 (phantom-brand persistence) from v0.6. Bed Bath & Beyond, ~37 months pre-measurement, is the cleanest available test of whether the v0.6 Mint observation generalizes.

The brand registry comprises 21 retailers spanning three tiers. Seven incumbents (Target, Amazon, Walmart, Costco, IKEA, HomeGoods, Wayfair) plus the Williams-Sonoma Inc. corporate-portfolio mid-tier (Williams-Sonoma, Pottery Barn, West Elm) plus specialty retailers (Crate & Barrel, World Market, Sur La Table, Container Store, At Home), four DTC challengers (Brooklinen, Parachute, Boll & Branch, Quince), the phantom test subject (Bed Bath & Beyond), and a phantom comparator (Pier 1, which collapsed in 2020). Boundary excludes pure furniture retailers, marketplaces, and grocery — preserving BBB's pre-collapse cognitive slot of 'where consumers go specifically for home essentials.'

Six prompts, each anchored to a Category Entry Point per Protocol §3.1: **FUNCTIONAL_WHY** ('best store to buy household essentials'), **CONTEXTUAL_WHEN** ('wedding registry retailers'), **CONSTRAINT_WITH** ('affordable furnishing on a tight budget'), **IDENTITY_HOW_FEELING** ('stores people actually shop at when setting up a household'), **DISCOVERY** ('emerging or innovative home goods brands'), and **COMPARISON** ('compare leading household goods retailers'). None named BBB or Pier 1 — that was the v0.6 design error in the personal finance category, where naming Mint by name compromised that prompt's phantom signal.

Six frontier AI models from four labs spanned the production lineup as of May 2026: Anthropic Claude Sonnet 4.6 and Claude Opus 4.7 (within-Anthropic generational comparison), OpenAI gpt-5.4-mini and gpt-5.5 (within-OpenAI mini-vs-flagship comparison), Google Gemini 2.5 Flash, and xAI Grok 4.1 Fast (with the oldest knowledge cutoff in the lineup at November 2024). Each prompt × model combination ran eight times at temperature 0.7 where supported. **288 successful measurements** after a pre-registered failed-call recovery cycle restored an initial 67% completion rate to 100%.

Brand mention extraction used gpt-5.4-mini at temperature 0 with structured-output classification, producing canonical brand mentions per response with rank, sentiment, and primary-recommendation flags. A second-stage manual review classified every BBB mention along two axes: a five-level valence taxonomy (**live_recommendation** / **live_with_caveat** /

status_correction / **historical_reference** / **ambiguous**) and a four-level entity-reference taxonomy (**legacy_brand** / **corporate_parent** / **both** / **unclear**). 110 BBB mentions reviewed; an independent spot-check audit on a stratified 25-row sample produced **88% strict agreement** with the AI classifier, with all three boundary disagreements clustering on the caveated/correction/historical adjacency and zero disagreements on the headline naive-phantom labels.

Eight hypotheses pre-registered before measurement: H1 phantom Presence ≥ 25%; H2 cross-model spread ≥ 25 points; H3 Sonnet > Opus by ≥ 15 points (within-Anthropic freshness); H4 within-OpenAI freshness as descriptive; H5 Grok ≥ Sonnet on phantom rate (oldest-cutoff comparator); H6 BBB highest in p2/p6, lowest in p5; H7 Pier 1 < BBB AND Pier 1 < 15%; H8 Beyond, Inc. < 25% of BBB+Beyond mentions. Pre-registration document locked at **PRE_REGISTRATION_household_v1.0.md** on 2026-05-04 before any data collection.

FINDING 01

The phantom replicates

Bed Bath & Beyond's aggregate raw AI Presence across all 288 measurements is **38.2%** — within four percentage points of Mint's 44% in v0.6 personal finance. The pre-registered H1 threshold of $\geq 25\%$ is cleared by 13 percentage points, in a category where the brand has not operated as a physical retailer for 37 months.

The result holds across model providers. Of the six models in the lineup, all six surfaced BBB at meaningfully non-zero rates. Anthropic Opus 4.7 at 56.2%, Anthropic Sonnet 4.6 at 41.7%, OpenAI gpt-5.4-mini at 50.0%, xAI Grok 4.1 Fast at 50.0%. Even the lowest two — Google Gemini 2.5 Flash at 16.7% and OpenAI gpt-5.5 at 14.6% — surfaced BBB at rates substantially higher than the H1 floor.

The household-goods leaderboard places BBB at #11 of 21 registered retailers, between Williams-Sonoma (36.8%) and West Elm (28.8%). The brand surfaces at rates above 13 of 21 registered live retailers, including Brooklinen, Parachute, Container Store, Boll & Branch, Quince, and five others currently operating in the category.

Pier 1 — the phantom comparator added explicitly to test whether pre-collapse footprint size and years-since-collapse are separable variables — surfaced in **0 of 288 measurements**. Zero. Across every model, every prompt, every run.

The phantom hypothesis is sharpened by this contrast: BBB at 38.2% and Pier 1 at 0.0% in the same registry, against the same prompts, across the same models. Two retailers with structurally similar fates (collapse + online-only revival) but different pre-collapse footprints and different years-since-collapse produce dramatically different phantom signals.

We cannot separate the two variables from a single measurement. Both contribute. But we now know a brand's pre-collapse cumulative editorial weight matters for phantom persistence — and a smaller pre-collapse footprint with longer time-since-collapse drops the phantom signal to zero. The Mint v0.6 observation could not pull this lever; the inclusion of Pier 1 in this design pays off. **H7 is confirmed in the strongest possible form.**

Two phantoms, two different fates

Bed Bath & Beyond and Pier 1 share a fate (collapse → online-only revival) but differ on years-since-collapse and pre-collapse footprint. The phantom signal between them differs by 38.2 percentage points.

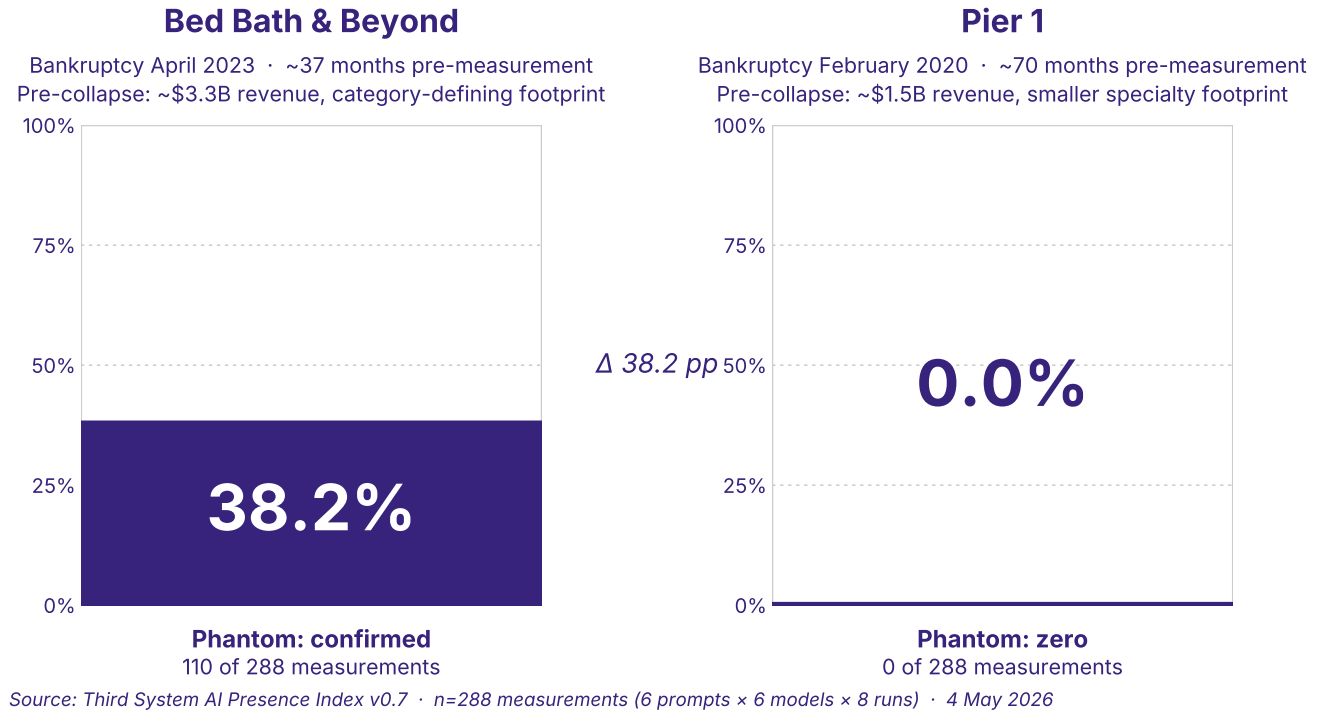


Figure 1 · BBB and Pier 1 share a fate (collapse → online-only revival) but differ on years-since-collapse and pre-collapse footprint. BBB at 38.2% phantom, Pier 1 at exactly 0.0% across all 288 measurements.

FINDING 02

The phantom is not a knowledge gap. It is a recommendation slot.

Of 110 BBB mentions across 288 measurements, the valence breakdown is: **5 live_recommendation** (4.5% of BBB mentions, 1.7% of all measurements — AI presents BBB as fully live with no caveat); **72 live_with_caveat** (65.5% of BBB mentions, 25.0% of all measurements — AI surfaces BBB with disclosure of the closure or rebrand); **7 status_correction** (6.4% / 2.4% — AI explicitly says BBB is closed and does not recommend it); **25 historical_reference** (22.7% / 8.7% — AI mentions BBB only in past tense as a former retailer).

Read this carefully. The headline number that drove the v0.6 Mint finding — and that drove the H1 confirmation in this measurement — is the raw mention rate, ignoring valence. **At the valence level, the picture inverts.** AI is not blindly recommending BBB as a live retailer. AI is mostly saying 'Bed Bath & Beyond closed its physical

stores in 2023, but bedbathandbeyond.com still operates online' — surfacing the brand alongside acknowledgment of its closure.

In **88% of BBB mentions**, AI demonstrates knowledge that the entity has changed. The naive-phantom rate — AI presenting BBB as fully live with no caveat — is 1.7% of all measurements. That is the actual answer to 'is AI telling consumers to shop at a defunct retailer?' But the brand still surfaces in 38.2% of responses. That gap — between AI's correct knowledge and AI's continued surfacing — is where the structural finding lives.

We propose a reframe: **phantom-as-recommendation-slot, not phantom-as-knowledge-gap**. The phantom in AI mediation is not a lag in what the AI knows. The phantom is in what the AI's recommendation set is structured to fill. Bed Bath & Beyond's accumulated editorial volume in AI training corpora — decades of

comparison articles, registry guides, household-essentials listicles, store-by-store rankings — encoded a recommendation slot. The slot persists in AI's recommendation logic regardless of the entity's current operating status. When the entity changes, AI does not vacate the slot. AI fills the slot with disclaimer.

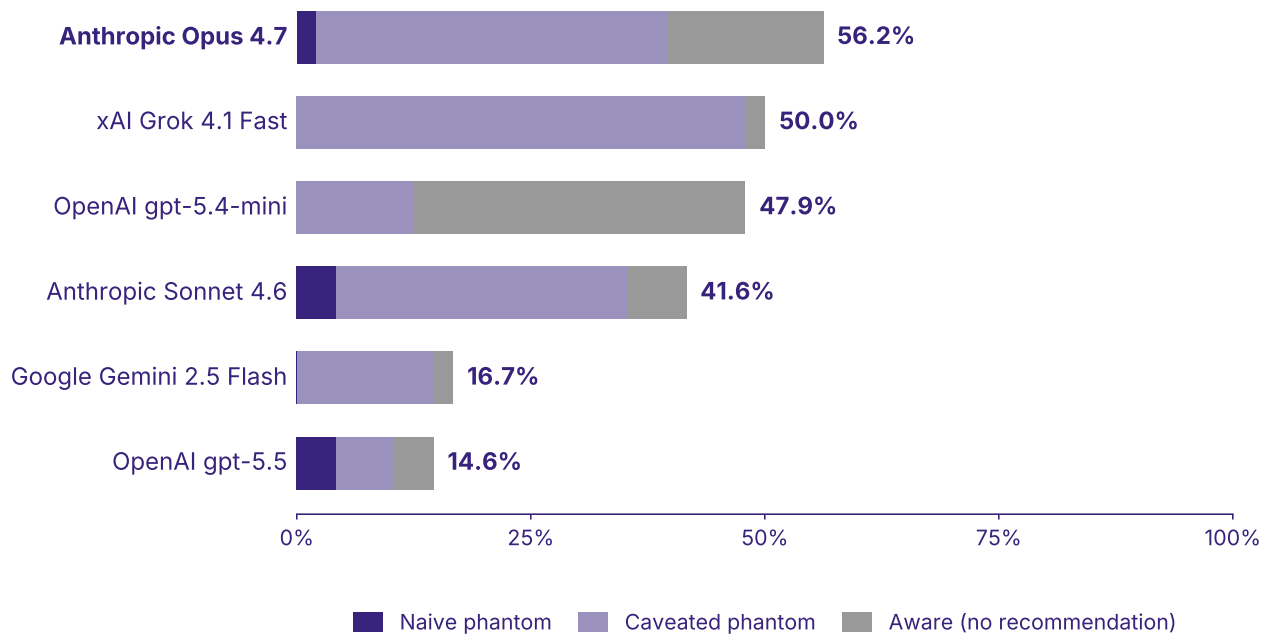
The implication for brand strategy is direct. AI-mediated visibility outlasts entity integrity in disrupted categories. Brands that have collapsed, rebranded, or merged continue to occupy recommendation slots that consumers encounter — with disclosure, but encounter. Competitors entering disrupted categories may find their recommendation slot already partially occupied by the structural ghost of the former leader. And the

phantom slot is most filled when the prompt activates a frame the brand was strongly editorially associated with — registry, household setup, kitchen basics — even when the AI's underlying knowledge of the entity is up to date.

This is a stronger structural claim than naive lag. Naive lag suggests training-data freshness as the variable to optimize. Recommendation-slot persistence suggests brand-mention pathways are reinforcement-encoded structures that retraining does not directly address. The same brand can live in different recommendation slots across different models — and across different prompts within the same model — even when every model's underlying knowledge of the entity is identical.

How each model handles Bed Bath & Beyond

Per-model valence breakdown of BBB mentions (denominator: 48 measurements per model).



Source: Third System AI Presence Index v0.7 · n=288 measurements (6 prompts × 6 models × 8 runs) · 4 May 2026

Figure 2 · Per-model valence breakdown of BBB mentions. Naive phantom is 1.7% of all measurements; caveated phantom is 25.0%. AI knows the entity has changed in 88% of mentions — it surfaces the brand anyway, with disclosure.

FINDING 03

Newer models phantom-mention better, not less

The pre-registered H3 — Anthropic Sonnet 4.6 BBB Presence > Anthropic Opus 4.7 BBB Presence by ≥ 15 points — was

disconfirmed at the raw-Presence level. Opus surfaces BBB more often than Sonnet (56.2% vs 41.7%), reversing the predicted

direction by 14.5 points.

The valence breakdown rescues the underlying hypothesis at a different level. Of Opus's 27 BBB mentions, 96.3% are handled responsibly (caveated phantom + aware mode). Of Sonnet's 20 BBB mentions, 90.0% are handled responsibly. Opus surfaces BBB more often, but a higher fraction of its mentions are in the aware-mode cluster (status correction or historical reference) rather than the recommendation cluster.

This pattern requires a more careful claim. **Newer models do not necessarily phantom-mention less. They phantom-mention better.** The freshness/reasoning differential between Sonnet 4.6 and Opus 4.7 shows up at the handling level — Opus is more likely to disclose the closure or historicize the mention — not at the frequency level. The recommendation slot persists; the disclosure improves.

The within-OpenAI comparison (gpt-5.4-mini vs gpt-5.5, H4 reported descriptively) reveals the same pattern. gpt-5.5 surfaces BBB at the lowest rate of any model in the lineup (14.6%) — the freshness-decay hypothesis would predict this is the cleanest

model. But of those 7 mentions, 4.2% are naive-phantom and 6.2% are caveated. By contrast gpt-5.4-mini surfaces BBB at 50.0%, with **35.4% in aware mode** — the highest aware rate of any slot in the dataset. gpt-5.4-mini, the older and smaller OpenAI model, handles BBB more correctly per-mention than the newer flagship.

The cross-lab pattern confirms H5: Grok 4.1 Fast — with the oldest knowledge cutoff in the lineup (November 2024) — produces phantom Presence at 50.0%, edging Sonnet 4.6 at 41.7% by 8.3 points. But Grok's pattern is its own structural type. Grok's caveated-phantom rate (47.9%) is the highest of any model. Grok's aware rate (2.1%) is among the lowest. **Grok almost always recommends BBB, almost always with caveat.** Of Grok's BBB mentions, 58.3% explicitly reference both entity names — far higher than any other model. The mechanism: Grok's older training cutoff captures the active 2023–2024 discourse around BBB's bankruptcy, the rebrand to Beyond, Inc., and the Overstock acquisition, locking in a 'BBB-as-active-with-disclosure' pattern that newer models with later training data treat as more historicized.

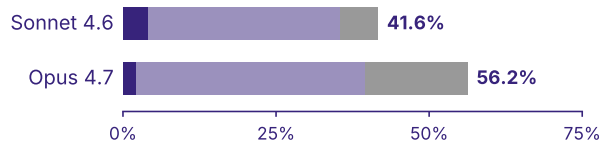
This is a finding about how AI ecosystems differ. Anthropic models surface BBB at high rates with mostly responsible handling. Grok surfaces BBB at high rates with always-disclosed handling. OpenAI mini is the most aware model, handling BBB as past-tense in a third of mentions. OpenAI flagship and Gemini Flash mention BBB rarely but inconsistently when they do. **Each lab's training pipeline produces its own phantom signature** — a structural property of the lab, not just the model.

Within-lab freshness

Newer = more responsible, not less frequent

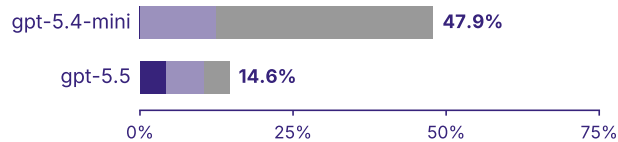
Anthropic

Sonnet 4.6 → Opus 4.7



OpenAI

mini → 5.5 (newer)



Source: AIPI v0.7 · 4 May 2026

Figure 3.

FINDING 04

Different prompts activate different temporal frames

The per-CEP analysis reveals a finding outside the pre-registration: phantom valence varies systematically with the cognitive frame the prompt activates. Same brand, same models, six different prompts — and BBB occupies different temporal slots depending on what the prompt asks.

In FUNCTIONAL_WHY prompts ('best store for bedding, towels, kitchen basics'), BBB raw Presence is 70.8%, and 60.4% of

measurements produce caveated phantom recommendations — AI surfacing BBB-as-still-relevant-online-retailer with disclosure of the physical-store closure. **Functional queries get BBB-as-current.**

In IDENTITY_HOW_FEELING prompts ('stores people actually shop at when setting up a household'), BBB raw Presence is 81.2% — the highest of any prompt — but 43.8% of measurements

produce aware-mode mentions (historical references or corrections). **Identity queries get BBB-as-cultural-memory.**

This is a temporal cognitive geometry the v0.6 framework did not anticipate. **BBB lives in different prompt frames as different temporal entities.** When AI is asked functionally, BBB surfaces as a current online retailer with disclosure. When AI is asked about cultural identity around home setup, BBB surfaces as a former retailer in past tense. Same brand, different prompts, different cognitive-time slots.

The implication for brand strategy is subtle and worth attention. A brand's AI-mediated visibility is not single-valued, even within a category. The same brand can simultaneously occupy a current-recommendation slot in one prompt frame and a historical-anchor slot in another — and consumers querying

different frames will encounter materially different brand presentations. For disrupted brands, the cognitive geometry creates a particular pattern: continued recommendation in functional frames, continued historical anchoring in identity frames, neither in comparison frames where live competitors saturate the surface.

H6 is partially confirmed: lowest in DISCOVERY (0.0%, exactly as predicted) but highest in IDENTITY and FUNCTIONAL (not in CONTEXTUAL or COMPARISON as pre-registered). The COMPARISON-frame absence is explained by Risk #2 from the design document — Walmart, Amazon, Target, and IKEA saturate the comparison frame at near-100% each, leaving structurally no room for BBB to surface in that prompt regardless of phantom strength.

The same brand, different temporal frames

Per-prompt valence breakdown of BBB mentions. Functional prompts activate BBB-as-current-online-retailer. Identity prompts activate BBB-as-cultural-memory. Same brand, six prompts, six different cognitive-time slots.

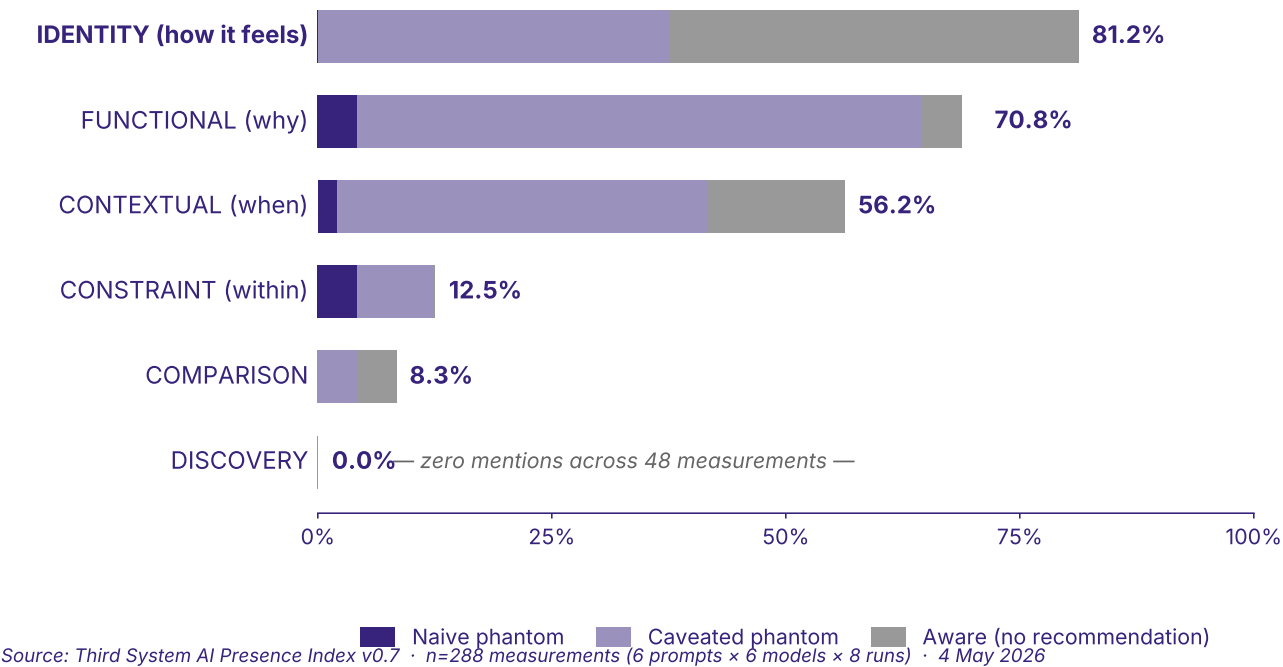


Figure 4 · Per-CEP temporal frame breakdown. Functional prompts produce BBB-as-current-online-retailer (60.4% caveated phantom). Identity prompts produce BBB-as-cultural-memory (43.8% aware-mode). Discovery prompts produce zero BBB mentions.

FINDING 05

The rebrand has not propagated. The legacy brand has.

H8 is confirmed: of 110 BBB mentions, 0% reference Beyond, Inc. as a corporate parent in isolation, 23.6% reference both names with awareness they are related, 75.5% reference only the legacy 'Bed Bath & Beyond' string. By either count — strict or inclusive — Beyond, Inc. is below the 25% pre-registration threshold.

The cross-tabulation reveals the structure. Of the 5 naive-phantom mentions (live_recommendation), **100% reference only the legacy name.** AI in naive-recommendation mode has not internalized the rebrand. Of the 72 caveated-phantom mentions, 65.3% reference only the legacy name and 34.7% reference both — caveated mentions are where the rebrand information lives, used to construct the caveat itself.

The mechanism is consistent: when AI knows about the rebrand, the rebrand information is deployed as part of the disclosure ('now operates as Beyond, Inc.'), not as a replacement of the legacy name in AI's recommendation set. The recommendation set remains anchored on 'Bed Bath & Beyond.' Beyond, Inc. lives as a

footnote.

For a marketing strategist, this has direct consequences.

Rebranding a disrupted entity into a new corporate identity does not propagate into AI's recommendation surface at the same rate that the legacy brand persists. The new entity name occupies a footnote slot, not a recommendation slot. The economics of rebranding in the AI mediation era include this asymmetry: the legacy brand continues to drive AI visibility long after the corporate identity has changed, and the new identity captures only the disclosure-language slot.

This finding is specific to a particular rebrand pattern (distressed parent acquired, online-only revival under legacy domain) and may not generalize to other rebrand types — organic name changes for live brands, acquisition rebrands without distress, holding-company consolidation. Phase 3 will measure additional rebrand patterns to test how broadly this asymmetry holds.

Limitations

Phase 2 measures one designed-for-test category. The findings below qualify the strength of the result. Six caveats apply.

Single category. Phase 2 measures one designed-for-test category. The phantom pattern observed for BBB cannot be generalized to all brand disruptions on the basis of this measurement plus v0.6's Mint observation. The replication moves the program from one-category claim to two-category claim — substantively defensible, not yet generalizable.

Inter-rater agreement on boundary cases. Manual review classifications used a single AI classifier (gpt-5.4-mini at temperature 0). An independent spot-check audit on a stratified 25-row sample produced 88% strict agreement, with three boundary disagreements clustering on the caveated/correction/historical adjacency. The disagreements do not affect the headline naive-phantom rate (1.7%, all five live_recommendation rows verified correct on independent review) but introduce a small error budget on the aggregate aware-mode counts.

Walmart/Amazon ceiling compression. Risk #2 in the pre-registration design document predicted that COMPARISON and FUNCTIONAL prompts would be ceiling-compressed by Walmart and Amazon at near-100% each, leaving constrained headroom for specialty brands including BBB. The prediction held. BBB's COMPARISON-prompt rate of 8.3% partly reflects this constraint.

Registry leakage. The unknown-mentions list includes TJ Maxx (132 mentions) and Marshalls (129 mentions) as the highest unknowns — both TJX sister brands of HomeGoods, which is in the registry. Phase 3 measurements should add the TJX sister set to the registry. Threshold (Target's house brand) appears at 42 mentions, representing component-mode leakage rather than retailer-mode mentions; this does not affect BBB analysis.

Single point in time. Each measurement reflects AI behavior on May 4, 2026. The findings do not describe how AI brand visibility moves over time. Phase 2 specifically established phantom persistence at a single time point; longitudinal measurement will require periodic re-baselining. The phantom-as-recommendation-slot framing predicts that retraining alone may not vacate the recommendation slot; this prediction will be testable when the same models update.

Construct validity remains unproven. The AI Presence Index measures a real and stable property of the AI tier, but whether that property correlates with consumer consideration, purchase intent, or sales is currently unknown. The phantom finding is a property of

AI mediation, not of consumer behavior; whether consumers in disrupted categories actually act on AI's caveated recommendations of defunct brands is the separate question the AIAS Phase 3 construct-validity program will address.

What's next

analysis. Brand strategy in the AI-mediated era requires measurement that can be inspected.

Phase 3 — longitudinal & construct validity. The recommendation-slot framework, if it generalizes, predicts that retraining will not vacate phantom slots — only redistribute the disclosure language used to fill them. A controlled test would measure the same disrupted brand on the same model across a major model update, comparing phantom Presence and valence distribution before and after. This is a direct test of the lag hypothesis we are now reframing.

Phase 3 will also correlate AI Presence and valence distribution against external consumer-tracking data. For BBB specifically, Google Trends, Statista consumer-awareness surveys, and DTC brand-tracking instruments are candidate validators. The phantom-as-recommendation-slot framing predicts that consumer awareness of BBB will be lower than AI Presence (the slot persists in AI even when consumer attention has moved on). If the prediction holds, AI Presence is a lagging indicator in disrupted categories; if it inverts, AI Presence is leading consumer behavior. Both findings are publishable; neither is established.

Methodology paper. The within-lab handling pattern (newer/smaller models showing different valence distributions than newer/flagship models within the same lab) suggests that AI handling of disrupted brands is not a single-variable function of training cutoff or model size, but interacts with reasoning-mode and instruction-tuning practices. A measurement isolating model-size-vs-cutoff (running the same brand against same-cutoff models of different sizes within a lab) would clarify the mechanism.

A methodology paper using this dataset as its empirical spine is planned for the AIAS measurement program's next publication cycle. The paper will present the v0.6 cross-category findings and the Phase 2 BBB measurement as a coherent empirical foundation for the AI Availability Score framework introduced in **Tri-System Brand Growth** (Ulpiano Gonzalez Castro, 2026), with the recommendation-slot reframe as the connective thesis.

Phase 4 — remaining AIAS components. Phase 4, contingent on Phase 3 results, will release the remaining five AIAS components: Ranking, Consistency, Coverage, Grounding, and Sentiment. The Phase 2 BBB measurement provides preliminary data on Sentiment and Coverage as a side benefit of the manual-review process — preliminary results will be folded into Phase 4 documentation when those components ship.

Third System publishes its methodology, its data, and its corrections openly. Each subsequent report includes the underlying response dataset alongside the headline findings, allowing critics, peer reviewers, and customers to verify the

Hypothesis scoring

Eight hypotheses locked in **PRE_REGISTRATION_household_v1.0.md** on 2026-05-04 before any measurement. Five confirmed, one partially confirmed, one disconfirmed at the level it was measured but supported at a different level, one descriptive.

H	Pre-registered prediction	Result	Status
H1	BBB Presence ≥ 25%	38.2% raw · 26.7% naive+caveated · 1.7% naive	Confirmed
H2	Cross-model spread ≥ 25 pts	41.6 pts (Opus 56.2 → flagship 14.6)	Confirmed
H3	Sonnet > Opus by ≥ 15 pts	Opus > Sonnet by 14.5 pts (raw); Opus more aware per-mention	Disconfirmed at raw, supported at handling
H4	(descriptive only)	mini > flagship by 35.4 pts (raw); mini more aware per-mention	Descriptive: opposite directions raw vs handling
H5	Grok ≥ Sonnet	Grok 50.0 > Sonnet 41.7 by 8.3 pts	Confirmed
H6	Highest in p2/p6, lowest in p5	Highest in p4 (81.2)/p1 (70.8); lowest in p5 (0.0)	Partially confirmed
H7	Pier 1 < BBB AND Pier 1 < 15%	Pier 1 = 0.0% across all measurements	Confirmed (strongest form)
H8	Beyond, Inc. < 25% of BBB+Beyond	0% strict · 23.6% inclusive	Confirmed

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v0.3). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Ulpiano Gonzalez Castro, P. (2026). **Phantom Brand Persistence: AI Presence Index v0.7 — Bed Bath & Beyond, designed for test**. Third System. thirdsystem.ai/v07-phantom-persistence

Underlying datasets

results_v2_household_v1.0_final.csv · 288 measurements

results_enriched_household_v1.0_*.csv · brand extraction

manual_review_bbb_*.csv · 110 BBB mentions classified

presence_index_v0.3_household_v1.0_*.csv · per-brand Presence

cross_tab_valence_*.csv · valence × slot, valence × CEP, entity × slot

spot_check_audit_*.csv · 25-row independent review sample

Methodology log: PRE_REGISTRATION_household_v1.0.md (locked 2026-05-04).

Inquiries: hello@thirdsystem.ai